# Zelin Li

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## Education

**University of Minnesota, Twin Cities**, Bachelor's Degree in Computer Science — Sept 2025 – May 2027

- **GPA: 4.0/4.0**

**Beijing University of Posts and Telecommunications (BUPT)**, Artificial Intelligence — Sept 2023 – June 2025

- GPA: 3.85/4.0, **Rank: 1/180**

## Research Interests

LLM Inference and Post-Training · Zeroth-Order Optimization · Algorithm–System Co-Design · Optimization Theory

## Publications & Preprints

**Inference-Native Zeroth-Order Optimization.**
**Zelin Li**, Caiwen Ding.
arXiv:2605.28760, 2026.

**Sample-Stable Subspaces and Mini-Batch Noise in Zeroth-Order Fine-Tuning.**
**Zelin Li**, Caiwen Ding.
Preprint, 2026.

**DynamicMem: A Long-Horizon Memory Benchmark in Real-World Settings.**
Wenya Xie, Shengming Zhou, **Zelin Li**, et al.
arXiv:2606.22877, 2026. Under review at NeurIPS 2026.

## Research Experience

**Research Assistant, University of Minnesota, Twin Cities** — Apr 2026 – Present
*Advisor: Prof. Caiwen Ding*

**Inference-Native ZO Systems**

- Formulated ZO as candidate-state queries and developed an inference-native execution path that separates algorithmic probe semantics from runtime realization.
- Built the vLLM-based prototype with LoRA-side candidate/update state, lazy and banked updates, packed variable-length scoring, and mixed foreground inference/ZO execution.

**ZO Stability and Mini-Batch Noise**

- Studied why structured, low-dimensional ZO remains batch-sensitive even after the search geometry becomes highly stable across samples.
- Designed large-batch, fixed-subspace, and matched-direction experiments on OPT-13B and Qwen3 to isolate mini-batch objective noise from search-space variation.

**Research Assistant, University of Minnesota, Twin Cities** — Nov 2025 – Mar 2026
*Advisor: Prof. Zirui Liu; worked with Wenya Xie*

**Long-Horizon LLM Memory Benchmark**

- Implemented and evaluated HippoRAG and SimpleMem; contributed to benchmark-data construction and extensive manual auditing of generated data.
- Used audit findings to support data and algorithm revisions, and contributed to experimental analysis and manuscript preparation.

## Honors & Awards

**National Scholarship**, Ministry of Education of China (5/470) — 2024
**Silver Medal**, International Collegiate Programming Contest (ICPC), Asia Shenyang Regional (84/399) — 2024
**Bronze Medal**, China Collegiate Programming Contest (CCPC), Harbin — 2024
**Bronze Medal**, China Collegiate Programming Contest (CCPC), Beijing — 2024
**First Prize**, National Olympiad in Informatics in Provinces (NOIP), Shanxi — 2021

## Technical Skills

**Programming:** Python, C++ **Frameworks:** PyTorch, vLLM, Hugging Face Transformers **Tools:** Linux, Git